# Estimating hidden post views from reactions: transferring a model across platforms

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## 1 Introduction

Media monitoring companies such as Brand24 report to their clients how many people may have seen a mention of their brand. The simplest reach measure, the view count, is often not visible: platforms limit what they show publicly, and the scope of this data differs between platforms and post formats and changes over time. In practice reach is therefore approximated by a fixed multiple of followers or reactions, which says nothing about how wrong it may be.

Predicting content popularity is widely studied [1], also across services: Shulman et al. [2] transfer a model between music, book, photo and link datasets, and Roy et al. [3] transfer knowledge from Twitter to YouTube videos. These works, however, predict future popularity and have popularity data from the target platform on which the model is trained or calibrated. Our task is different: we estimate current views from visible reactions on a platform from which the model gets not a single view count in training. We collect views from that platform separately and use them only for evaluation. We found no work that tests transferring a view model across platforms in this way.

We ask whether the views of a Facebook or Instagram post can be estimated from its reactions by a model trained only on other platforms, and with what uncertainty. Our contributions:

- we show that the ratio of views to likes, estimated only on X and YouTube, transfers to other platforms: the model estimates the views of Facebook videos with a typical error (factor $10^{\mathrm{MAE}}$, Section 3) of $\times 2.43$, the same as a model trained on Facebook itself, and of Instagram reels with $\times 2.72$ (model trained on Instagram: $\times 2.41$); the simplest rule that knows no views from the target platform is off by $\times 9$–$\times 20$;
- we show when this transfer works (reactions read together with views) and when it fails (X, Stack Exchange);
- we give an uncertainty interval for every post, on the same platform valid also when data are missing;
- we show applications: a breakout alert, a brand ranking and media value.

## 2 Data

We use 12 500 mentions of 348 brands from five platforms, provided by Brand24 (2 500 per platform; 85% were published between April and August 2026, and 85% are in English). For TikTok, YouTube and X the views are in the export. For Facebook and Instagram they are not, so we read them in November 2026 from the public post pages, without logging in, together with likes and comments from the same moment. On Facebook a script read the view counter of the video player (5 November): 402 of 470 videos (68 removed or unavailable) from 20 of 22 brands. On Instagram a browser agent read the reel pages (6–8 November): of 976 reels, 261 were removed, 19 had no counter and 54 had hidden likes, leaving 642 reels from 11 brands. The Instagram sample thus covers only reels with visible likes. Ordinary posts on these platforms have no view counter, so the results apply only to videos and reels. Mention type and sentiment were labelled by language models and then checked and corrected by a team member.

**Table 1:** Data by platform.

| Platform | Brands | Authors | Source of views | Posts w/ views | Median views | Views/like |
|---|---|---|---|---|---|---|
| TikTok | 266 | 2 411 | Brand24 export | 2 491 | 729 | 25 |
| YouTube | 273 | 2 041 | Brand24 export | 2 441 | 433 | 26 |
| X | 252 | 1 658 | Brand24 export | 1 803 | 27 | 22 |
| Facebook | 20 | ≥182[a] | own read (videos) | 402 | 1 413 | 35 |
| Instagram | 11 | ?[b] | own read (reels) | 642 | 504 | 53 |

a: author known for 74% of videos. b: author known for only 11% of reels (59 authors among 72). Views/like: median ratio over posts with at least one like.

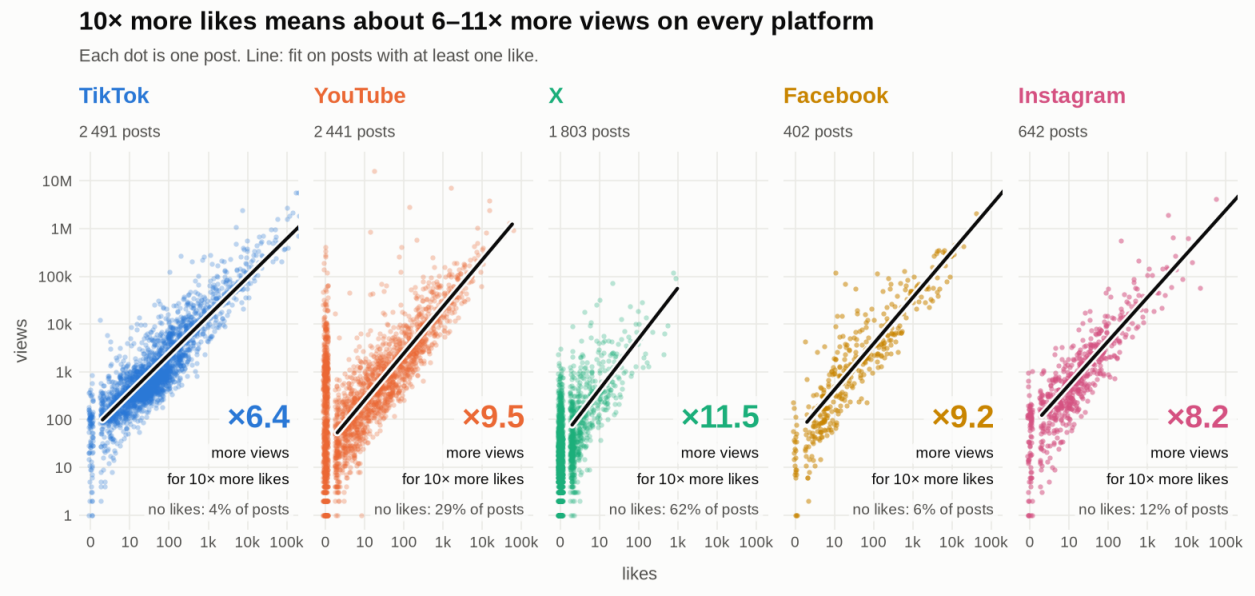



**Figure 1:** Views versus likes, separately for each platform. On every platform 10 times more likes means 6–11 times more views.
Both axes on a log scale; posts without likes at zero. Line: fit on posts with at least one like; number: how many times more views a post with 10 times more likes has. TikTok, YouTube, X: Brand24 export; Facebook (videos), Instagram (reels): our read.

# 3 Method

We predict $y = \log_{10}(\text{views})$. We report the error as MAE on this scale and as the *typical error*, the factor $10^{\text{MAE}}$ (MAE 0.30 $\approx \times 2$). Agreement in ordering is measured by Spearman's rank correlation between estimated and true views. Every model is compared with two rules: the median views and a *fixed likes multiplier* (views $= k \cdot$likes, $k$ the median ratio of views to likes in the training data, about 25). The gain over the median is reported as $1 - \text{MAE}/\text{MAE}_{\text{median}}$, because platforms differ greatly in the spread of views. The final model is evaluated with five-fold cross-validation in which the same brand, the same author or an identical text never appears in both training and test: one author may write about several brands (1–9% of posts), and the same text is sometimes pasted on different accounts. Every evaluation is thus on a brand and an author the model has not seen. Confidence intervals come from a bootstrap over brands. Method comparisons (Section 6 and Fig. 5B, same platform only) use a simpler split by brand alone, so their numbers differ from Fig. 2 by 0.01–0.03 (e.g. CatBoost on X: ×2.95 by brand against ×3.03 by brand, author and text); the ordering of methods does not change.

**Two scenarios, marked in every figure.** *Same platform*: the model is trained on posts from the given platform (TikTok, YouTube, X) and evaluated on brands and authors it has not seen. *Out of platform*: the model sees not a single post from the target platform in training (Facebook, Instagram, Stack Exchange; in Table 2 each platform in turn), and its views are used only for evaluation.

On TikTok, YouTube and X we use CatBoost [4] on followers, reactions and missingness flags (a separate feature says whether a field was empty, because missing is not zero). For the transfer to Facebook and Instagram we use ridge regression on $\log(1 + \text{likes})$ and $\log(1 + \text{comments})$, the two fields every platform has, trained on X and YouTube together [6]; with the target platform left out it transfers better than CatBoost

and MLPs (by 0.014–0.062 MAE on each of the five platforms). Per-post 50% and 90% intervals come from conformalized quantile regression (CQR) [5]: a quantile model gives a preliminary interval, and its width is corrected on a held-out part of the training data so that coverage matches the target. For Facebook and Instagram all main numbers use the Brand24 export for X and YouTube; the variant with re-read source data (Section 5) is reported separately, because it was tested after part of the Instagram reels had been seen.

# 4 Results

**Main result.** Fig. 2 shows both scenarios. On TikTok, YouTube and X, where the model is trained on the same platform, the typical error is ×1.9, ×2.9 and ×3.0. A fixed likes multiplier gives ×2.2, ×4.1 and ×3.6 here, so CatBoost beats it on every platform (by 0.08–0.15 MAE, confidence intervals exclude zero). On Facebook and Instagram the model has not seen a single post from these platforms, and its error is ×2.43 (confidence interval ×2.05–×3.03) and ×2.72 (×2.49–×3.05). The median views of X and YouTube are off there by ×20 and ×9, and even the median of the target platform itself by ×11 and ×6. On Facebook this equals the same model trained on Facebook itself (×2.42; difference 0.001, interval −0.03–0.03). On Instagram it is clearly worse than training on Instagram alone (×2.41; difference 0.054, interval 0.021–0.093), and 72% of reels are underestimated. The transfer is in practice a transfer of the views-to-likes ratio: a fixed likes multiplier from X and YouTube gives ×2.42 on Facebook and ×2.94 on Instagram, 0.034 worse than ridge. The ordering of posts is preserved well (Spearman 0.91 and 0.86, and 0.80 and 0.83 within a single brand). A typical brand, however, has a larger error than the whole: the median over brands is ×2.7 on Facebook and ×2.9 on Instagram.

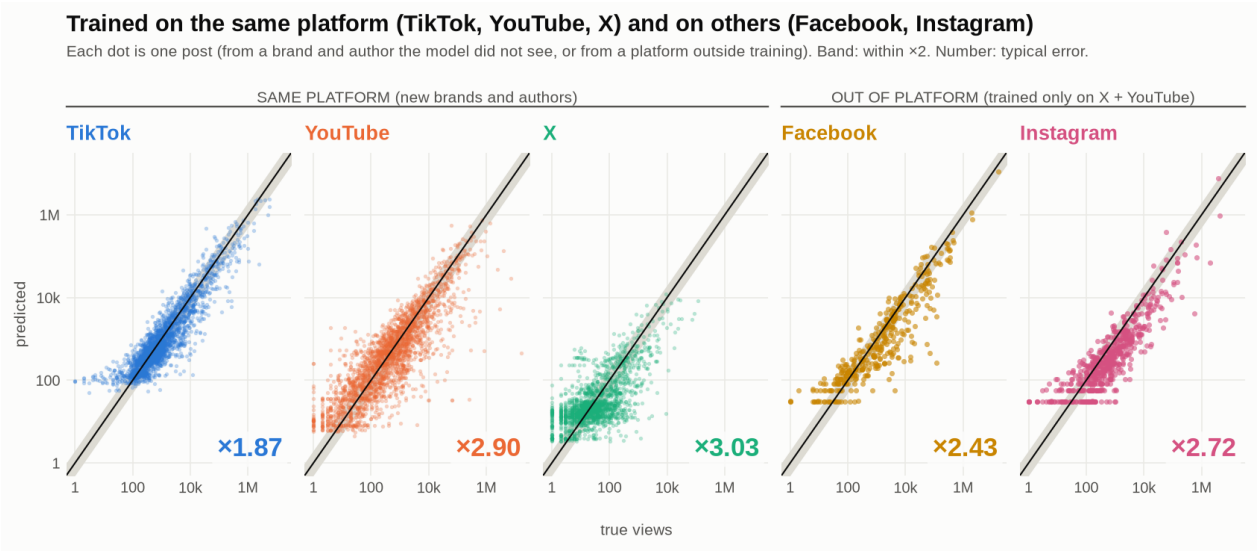



**Figure 2:** Estimated versus true views.
Each dot is one post. TikTok, YouTube, X: CatBoost on followers, reactions and missingness flags, trained on the same platform, cross-validation with no shared brands, authors or texts; truth from the Brand24 export. Facebook (402 videos, 20 brands), Instagram (642 reels, 11 brands): ridge on log likes and comments trained only on X and YouTube, without any post from the target platform; truth from our read. Grey band: estimate within ×2. Number: typical error $10^{\mathrm{MAE}}$.

**Who is estimated badly.** On every platform, also after transfer, the model overestimates posts with few views (about ×2 below 100 views) and underestimates large ones (Fig. 3). This is the usual regression to the mean: reactions do not separate a post that broke out from an ordinary one well. It is strongest on X, where posts above 10 000 views are underestimated about tenfold; after transfer, Instagram underestimates large reels by about ×3.

**Intervals.** On the same platform the 90% interval covers 87–91% of posts on TikTok, YouTube and X, also on months outside training (0.90–0.93). Out of platform, on Facebook, an interval calibrated on 25% of the Facebook brands covers 92% of videos: 94% of videos from accounts below one million followers, but only about 85% for the 70 videos from the largest accounts (±8 percentage points at this size).

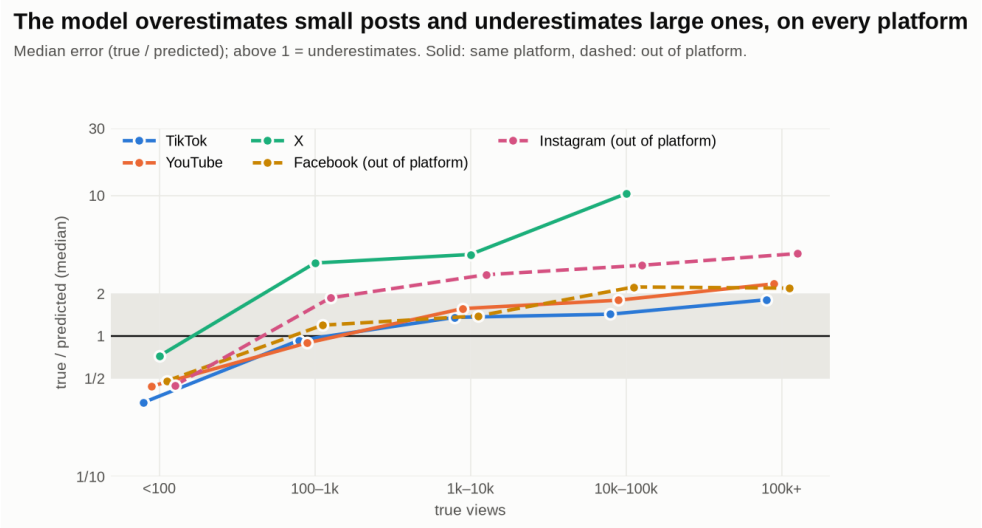



**Figure 3:** Error by post size.
Median ratio of true to estimated views in bins of true views (bins with at least 10 posts shown). Above 1: the model underestimates. Grey band: within ×2. Solid line: same platform, dashed: out of platform. Models as in Fig. 2.

# 5 Transfer across platforms

**Where we know the truth.** We have true views on all five platforms (Table 1): on TikTok, YouTube and X from the Brand24 export, on Facebook (videos) and Instagram (reels) from our read, together with likes and comments from the same moment. This lets us treat each platform as "hidden": we train the model on the others and test it on this one. We use only the two fields every platform has, likes and comments, and plain ridge regression, because with the target platform left out it transfers better than CatBoost. In practice Facebook and Instagram matter, since monitoring data have no views there; the other platforms show how general the result is.

**All combinations.** Table 2 shows the error for each of the 31 sets of training platforms. Three things stand out. First, Facebook and Instagram can be estimated from almost any set with an error of ×2.4–×3.1, close to a model trained on them alone (×2.4); training on X alone is the worst. Second, X is an exception in both directions: a model trained on X has an error of ×5.5 on YouTube, and no set gets below ×3.6 on X, against ×3.3 when trained on X. On X most posts have zero or one like (Fig. 1), so the relation of views to likes looks different there. Third, YouTube is hard even for a model trained on it (×4.1 for ridge against ×2.9 for CatBoost on all fields). This is mainly due to the export labels: on YouTube views re-read together with reactions, the same ridge from the other platforms is off by ×2.2. More platforms in training do not always help: what matters is which platforms are in the set and in what proportions. Confidence intervals (brand bootstrap) are about ±0.013–0.032 MAE on TikTok, YouTube and X, ±0.04 on Instagram and ±0.07 on Facebook, so differences between sets below about ×1.1–×1.2 on Facebook and Instagram are within noise.

**Three ranges instead of a number.** A report often only needs to say whether a post had fewer than a thousand, one to ten thousand, or more than ten thousand views. Fig. 4 shows this out of platform for each of the five platforms: a model trained on the other four, without any post from the target platform. Accuracy is 78–81%, but the ranges differ greatly in size, so balanced accuracy (mean recall over ranges) matters more: 78% on TikTok, 76% on Facebook, 70% on YouTube and 65% on Instagram, against 33% for guessing; macro F1 0.70–0.78. We are off by two ranges for 0.2–1.9% of posts. The middle range (1–10 thousand views, recall 50–73%) is usually the weakest; its posts are underestimated by one range. On Instagram the top range is the weakest (49%).

X looks different: accuracy 96%, but balanced accuracy only 46% and macro F1 0.52. On X 95% of posts have fewer than a thousand views (median 27), so a model that almost always says "below a thousand" gets high accuracy for free. Of 98 posts above a thousand it gets only 23 right. We see two reasons. First, 62% of posts on X have no like at all, so the model sees only a zero there and cannot tell a small post from a large one; removing such posts from the training data lowers the error on X by 0.08. Second, views in the X export are read early, about two times lower than in our re-read; on re-read X posts (partly different posts) the same model has an error of 0.49 instead of 0.64.

**Table 2:** Typical error (×) for all 31 sets of training platforms.
Ridge on log likes and comments, posts from the training platforms pooled into one set. Row: platforms in training (TT TikTok, YT YouTube, X, FB Facebook, IG Instagram), in groups of 1, 2, 3 and 4 platforms, best on average first within a group. Column: test platform, absent from training. First row: the same model trained on the test platform itself (brand-grouped CV). Bold: best result in the column. Darker: larger error.

| Trained on | TikTok | YouTube | X | Facebook | Instagram |
|---|---|---|---|---|---|
| *same platform (brand CV)* | 2.0 | 4.1 | 3.3 | 2.4 | 2.4 |
| YT | **2.0** | train | 4.6 | 2.7 | 2.6 |
| FB | 2.3 | 4.3 | **3.6** | train | **2.5** |
| IG | 2.5 | 4.5 | 4.1 | 2.5 | train |
| TT | train | 4.1 | 4.1 | 2.6 | 2.6 |
| X | 3.4 | 5.5 | train | 3.1 | 2.7 |
| YT + X | 2.1 | train | train | 2.4 | 2.7 |
| YT + FB | 2.0 | train | 4.5 | train | 2.5 |
| YT + IG | 2.0 | train | 4.6 | 2.6 | train |
| TT + X | train | 4.1 | train | 2.6 | 2.8 |
| TT + YT | train | train | 4.5 | 2.6 | 2.6 |
| X + FB | 2.5 | 4.5 | train | train | 2.7 |
| X + IG | 2.8 | 4.7 | train | 2.6 | train |
| FB + IG | 2.4 | 4.4 | 3.9 | train | train |
| TT + FB | train | **4.1** | 4.0 | train | 2.6 |
| TT + IG | train | 4.1 | 4.2 | 2.6 | train |
| YT + X + IG | 2.1 | train | train | **2.4** | train |
| YT + X + FB | 2.1 | train | train | train | 2.6 |
| TT + YT + X | train | train | train | 2.5 | 2.7 |
| YT + FB + IG | 2.0 | train | 4.4 | train | train |
| TT + X + IG | train | 4.1 | train | 2.5 | train |
| TT + YT + FB | train | train | 4.4 | train | 2.6 |
| TT + X + FB | train | 4.1 | train | train | 2.8 |
| TT + YT + IG | train | train | 4.4 | 2.6 | train |
| X + FB + IG | 2.6 | 4.5 | train | train | train |
| TT + FB + IG | train | 4.1 | 4.1 | train | train |
| YT + X + FB + IG | 2.1 | train | train | train | train |
| TT + YT + X + IG | train | train | train | 2.5 | train |
| TT + YT + X + FB | train | train | train | train | 2.7 |
| TT + X + FB + IG | train | 4.1 | train | train | train |
| TT + YT + FB + IG | train | train | 4.4 | train | train |

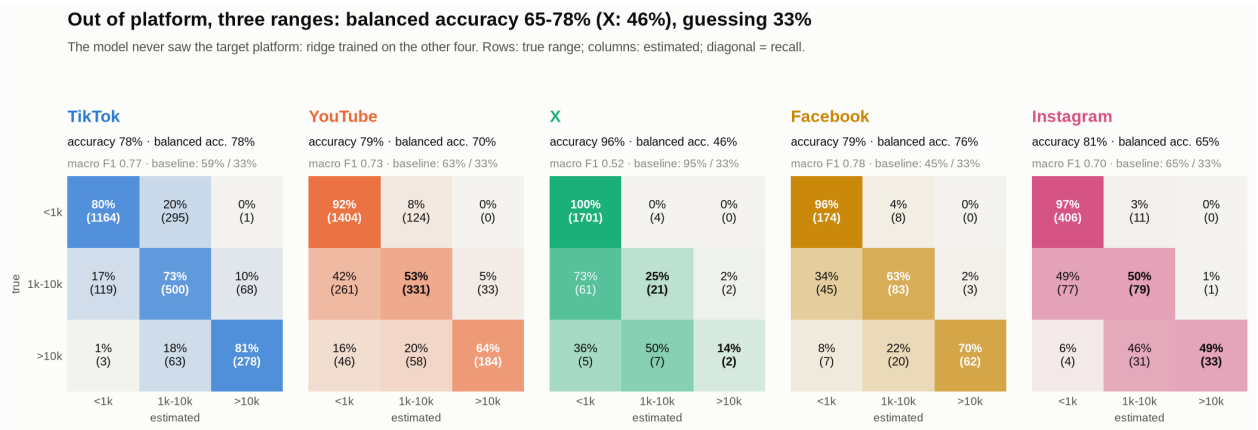



**Figure 4:** Out-of-platform estimate cut into three view ranges.
For each platform, ridge on likes and comments trained on the other four (as in the last row of Table 2). Row: true range; column: estimated; in a cell: share of the row and number of posts; the diagonal (bold) is the recall of the range. Above the table: accuracy, balanced accuracy (mean recall), macro F1; grey: accuracy and balanced accuracy when always guessing the most frequent range.

**What the model needs.** The transfer works only with likes read together with views (Fig. 5A). Comments alone give ×6.3–×7.4, and likes and comments from the export, read at a different moment than views, ×6.8. On platforms with known views (Fig. 5B) missing followers hardly hurt, while missing reactions raise the error to ×3.8–×5.2. For each pattern of available fields we train a separate model with its own interval, so the 90% interval covers 88–90% of posts in every case; a model trained on full data and fed incomplete data would cover only 46–83%.

**Fresh source data.** Views in the export are read shortly after publication, and our Facebook and Instagram reads several months later. When the same model is trained on re-read X and YouTube, with reactions from

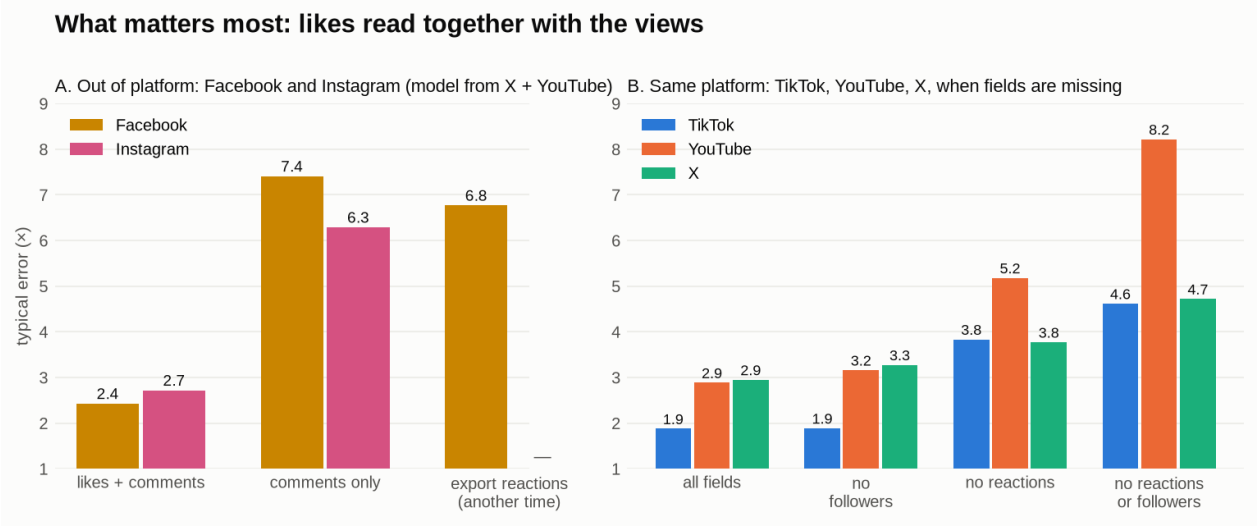



**Figure 5:** Typical error with different inputs.
A: Facebook and Instagram, ridge trained on X and YouTube. "Likes + comments": our fields read together with views; "comments only": a model on comments alone; "export reactions": likes and comments from the Brand24 export, read at a different moment than views (not available for Instagram). B: TikTok, YouTube, X, a separate CatBoost for each pattern of available fields, brand-grouped validation. A: out of platform, B: same platform.

the same moment, the error on Instagram drops from ×2.72 to ×2.41 (as much as a model trained on Instagram alone), and on Facebook it stays at ×2.44. The underestimation of large posts and of brand totals roughly halves. Merely shifting export views to a later age does not achieve this, so what matters is that views and reactions come from the same moment. This variant does not replace the main result, because the Instagram gain was checked after part of the reels had been seen.

**A sixth platform.** To test the transfer outside social media, we fetched through the public API 538 questions from Stack Exchange sites (among others Stack Overflow, Super User, Android, Apple, Mechanics) about 48 brands from our data, from the same period (April–August 2026). We treat an upvote as a like, and answers and comments as comments; this mapping was fixed before evaluation. Here the transfer fails. A model trained on X and YouTube is off by ×3.14, worse than a constant median from Stack Exchange itself (×1.96) and even than the median of X and YouTube (×2.10). 96% of questions are underestimated: question views accumulate over months and an upvote is not a like, so the whole level is shifted. The ordering transfers only moderately (Spearman 0.57, within brand 0.43). Question views also have a small spread (standard deviation 0.40 against 1.21 on Facebook), so even a ridge trained on Stack Exchange alone (×1.77) is only 15% better than the median. A fixed likes multiplier transfers even worse (×5.33). Conclusion: the views-to-reactions ratio transfers between social video services, but not to a question-and-answer site, where reactions mean something else.

# 6 Comparison of methods

**Model families.** Table 3 compares models on the same inputs. After tuning, almost all families give the same error: ×1.85–×1.92 on TikTok, ×2.8–×2.9 on YouTube and ×2.9–×3.1 on X. Only the simple rules (×2.2–×8.8) and linear regression and kNN on YouTube (×3.0–×3.5) are clearly worse. The best single model is TabPFN v2 [9], but it beats CatBoost by only 0.002–0.010, less than the width of the confidence interval (±0.012–0.021), and takes about 150 times longer. The graph network and a transformer reading the post as text are worse than CatBoost.

**Why not an ensemble.** An ensemble combines the estimates of several models, e.g. by averaging them ("mean of top 3") or with a second-level model that learns how to weight the estimates of all families ("stack"). Errors of different models partly cancel, so an ensemble is usually slightly better than each member. Here the stack improves the result by 0.011–0.028 (×1.03–×1.07). We keep a single model for three reasons: (1) the gain is small against the typical error of ×1.9–×3.0; (2) a stack requires training and maintaining over a dozen models, while the uncertainty intervals (CQR) and the transfer to other platforms are built on one model; (3) in the transfer the simplest linear model wins anyway (by 0.014–0.062 MAE, Section 3), so complexity does not carry over to the main result.

**Table 3:** Comparison of model families: MAE on the $\log_{10}$ scale and typical error (×).
All models on followers, reactions and missingness flags, brand-grouped cross-validation. Tree, linear, kernel and MLP models after hyperparameter tuning (40 settings per fold, nested tuning). TabPFN, TabICL, GraphSAGE, transformer and stack come from separate runs on the same data; CatBoost in those runs stayed within 0.271–0.273 / 0.452–0.456 / 0.466–0.470, so differences below about 0.005 are within noise. 95% confidence interval (brand bootstrap) for CatBoost: ±0.012 / ±0.021 / ±0.021. Bold: best single model in the column. Ensembles shown for comparison only. *: split with no shared brands, authors or texts.

| Model | TikTok | YouTube | X |
|---|---|---|---|
| *Simple baselines* | | | |
| platform median views | 0.663 (×4.60) | 0.944 (×8.79) | 0.667 (×4.65) |
| fixed multiple of followers | 0.919 (×8.30) | 0.855 (×7.16) | 0.879 (×7.57) |
| fixed likes multiplier* | 0.348 (×2.23) | 0.609 (×4.06) | 0.560 (×3.63) |
| *Linear and kernel models* | | | |
| linear regression (ridge) | 0.277 (×1.89) | 0.545 (×3.51) | 0.487 (×3.07) |
| k nearest neighbours (kNN) | 0.283 (×1.92) | 0.483 (×3.04) | 0.490 (×3.09) |
| SVR with RBF kernel | 0.271 (×1.87) | 0.463 (×2.90) | 0.480 (×3.02) |
| *Decision trees* | | | |
| random forest | 0.275 (×1.88) | 0.463 (×2.90) | 0.473 (×2.97) |
| extra trees | 0.275 (×1.88) | 0.459 (×2.88) | 0.475 (×2.99) |
| gradient boosting (scikit-learn) | 0.276 (×1.89) | 0.453 (×2.84) | 0.474 (×2.98) |
| LightGBM [8] | 0.275 (×1.88) | 0.457 (×2.86) | 0.475 (×2.99) |
| XGBoost [7] | 0.273 (×1.87) | 0.450 (×2.82) | 0.467 (×2.93) |
| CatBoost (final model) | 0.272 (×1.87) | 0.452 (×2.83) | 0.470 (×2.95) |
| *Neural networks* | | | |
| multilayer perceptron (MLP) | 0.276 (×1.89) | 0.454 (×2.84) | 0.477 (×3.00) |
| GraphSAGE [11] (similar-text graph) | 0.279 (×1.90) | 0.467 (×2.93) | 0.474 (×2.98) |
| fine-tuned transformer (row as text) | 0.289 (×1.95) | 0.454 (×2.84) | 0.490 (×3.09) |
| *Pretrained tabular models* | | | |
| TabPFN v2 | **0.268 (×1.85)** | **0.443 (×2.77)** | **0.465 (×2.92)** |
| TabICL [10] | 0.272 (×1.87) | 0.445 (×2.79) | 0.466 (×2.92) |
| *Ensembles (for comparison)* | | | |
| mean of top 3 families | 0.270 (×1.86) | 0.446 (×2.79) | 0.469 (×2.94) |
| stack of all families | 0.260 (×1.82) | 0.425 (×2.66) | 0.441 (×2.76) |

**Extra data.** We also tested more than twenty additional signal sources (Fig. 6): text (from TF-IDF to e5 sentence embeddings [15] and the VADER sentiment lexicon [16]), AI labels, author and brand graphs, and external brand data. None improves the error by more than ×1.04, most improve it by less than ×1.02, and some make it worse by up to ×1.06. Text helps slightly on YouTube and X, graphs and external data hurt. On Facebook text worsens the transfer (MAE 0.385 → 0.436). Author history is described separately below.

**More data without new labels.** We also tested methods that add data or learn its structure without views. Synthetic large posts (SMOTER/SMOGN [13]) reduce the underestimation of the most popular tenth of posts, but only by stretching the scale of the estimates, and raise the error by 0.006–0.019. Random perturbation along the trend and mixup do not let CatBoost or an MLP transfer as well as ridge (they stay 0.014–0.062 behind on each of the five platforms). A representation learned without labels (denoising autoencoder, SCARF [14]) changes the error by at most 0.005, and pseudo-labels for posts without views by at most 0.008. The final model therefore stays without augmentation.

**Author history.** The only signal that clearly helps is author history. The same author gets a similar number of views on successive posts, regardless of reactions: the correlation of the view surplus (truth minus estimate) between an author's first and later posts is 0.68–0.70 on all three platforms, and authors in the top third get 4–9 times more views than those in the bottom third at the same estimate. We use this simply: to the estimate of a new post we add the mean surplus of the author's earlier posts, shrunk towards zero when there are few posts (sum of surpluses divided by the number of posts plus a constant tuned on other brands). On posts by authors with history the error drops from 0.37 to 0.26 on TikTok, from 0.50 to 0.35 on YouTube and from 0.53 to 0.43 on X (×1.3–×1.4); most of the gain comes from a single earlier post. Such history, however, exists for only 3–16% of posts in our sample, so the gain vanishes in the mean error ($\leq$0.003). We also checked the same author on another platform. After normalising names (lower case, no special characters or suffixes such as "official") and handles from profile URLs, only 56 of 8 332 posts with views (0.67%; including re-read X and YouTube posts) have a post by the same author on another platform, so this signal cannot be used on monitoring data.

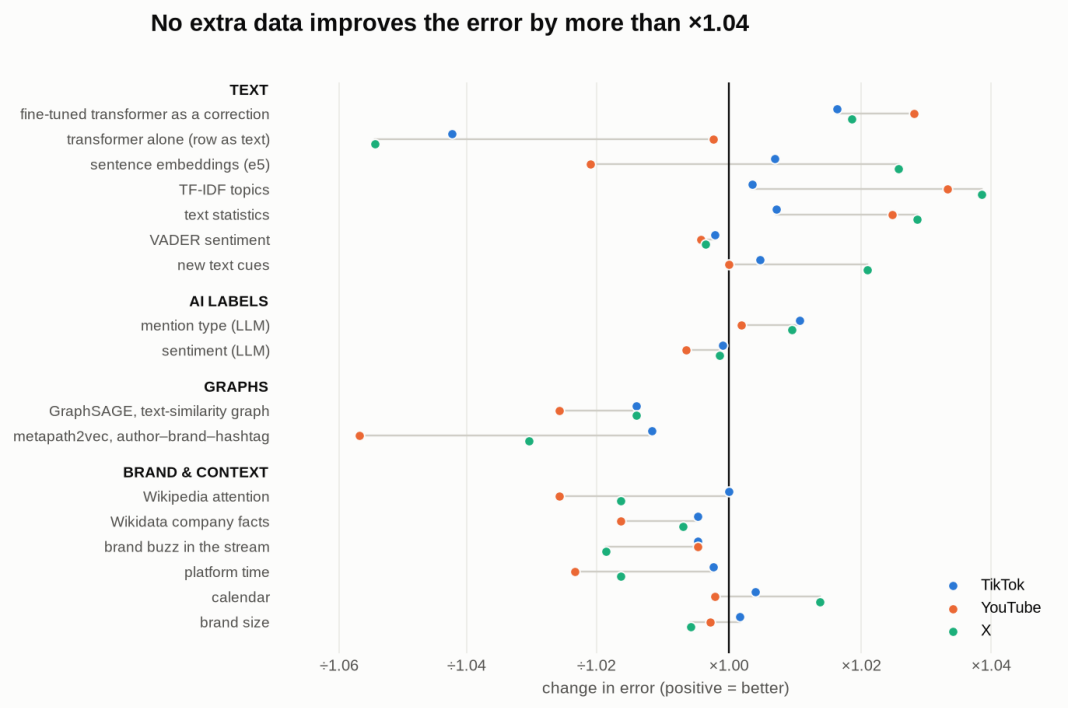



**Figure 6:** Extra data and features.
Change in typical error relative to the final model (transformer as correction, embeddings, graphs, external data) or relative to the model on followers and reactions (TF-IDF, text statistics, VADER, AI labels, calendar, brand size). Right: better, left: worse; ×1.02 is a change in MAE of about 0.01. *Transformer as correction*: a fine-tuned language model predicts the CatBoost error from the post text. *e5 embeddings*: a multilingual sentence model. *LLM labels*: mention type and sentiment from a language model. *GraphSAGE*: a graph network on the similar-text graph; *metapath2vec* [12]: embeddings of the author–brand–hashtag graph. *Wikipedia, Wikidata*: interest in the brand and company data. All with brand-grouped validation.

# 7 Applications

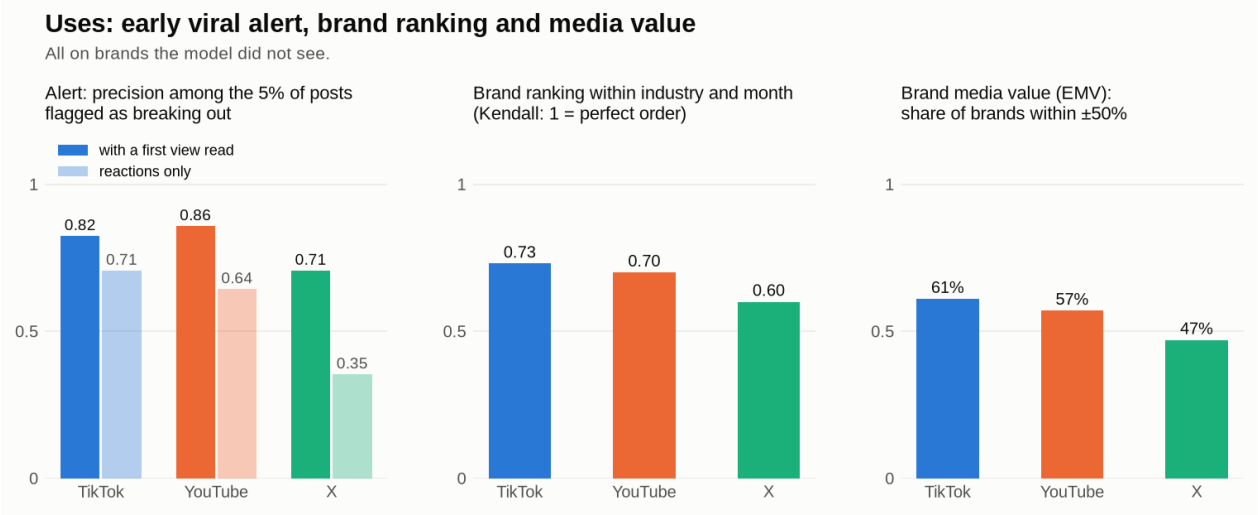



**Figure 7:** Applications of the estimate on TikTok, YouTube and X.
Alert: from posts read for the first time the model flags the 5% that will have the most views; the bar is the share of correct flags (289–336 posts per platform, one month). Ranking: mean Kendall coefficient between the estimated and true ordering of brands in industry–month pairs (52 / 51 / 42 pairs). EMV: views converted to money at a rate per thousand; share of brands whose estimated EMV is within ±50% of the true one (238 / 213 / 189 brands).

**Breakout alert.** With the first read of views the model flags the 5% of posts that will break out, with precision 0.82 / 0.86 / 0.71; reactions alone give 0.71 / 0.64 / 0.35 (Fig. 7). With three times more alerts it catches 82–100% of breakout posts.

**Brand ranking.** By summing the estimated views of brands within an industry and month we recover their order (Kendall 0.73 / 0.70 / 0.60).

**Media value.** A brand's EMV computed from estimated views is within ±50% of the true one for 61% / 57% / 47% of brands.

**Outlying posts.** On TikTok, posts with many more views than their reactions suggest are more often promotion (51% against 34%), and those with many fewer are more often spam (26% against 13%). **TODO:** fill to a full page: e.g. alert precision versus number of alerts, a ranking example for one industry.

# 8 Limitations and recommendations

The Facebook result rests on 20 brands and the Instagram result on 11; the three largest brands hold 39% and 44% of posts, and without them the error rises to ×2.89 and ×2.96. With so few brands the brand-bootstrap confidence intervals may be too narrow. On Instagram we measure only reels with visible likes. On the same platform the most popular 1% of posts on YouTube and X is underestimated ×9–×14, so brand totals are off by more than ×2 for 29–47% of brands (Fig. 8 shows that the reach column in the export is off even more). Almost every post has a different author (2 411 authors for 2 491 posts on TikTok, 2 041 for 2 441 on YouTube, 1 658 for 1 803 on X), so author history cannot be used for most posts, and on Instagram the author is usually unknown. The results describe associations, not causes. We recommend to the monitoring provider: read reactions together with views, record missing values as missing rather than zero, take a second read of views, and keep author histories.

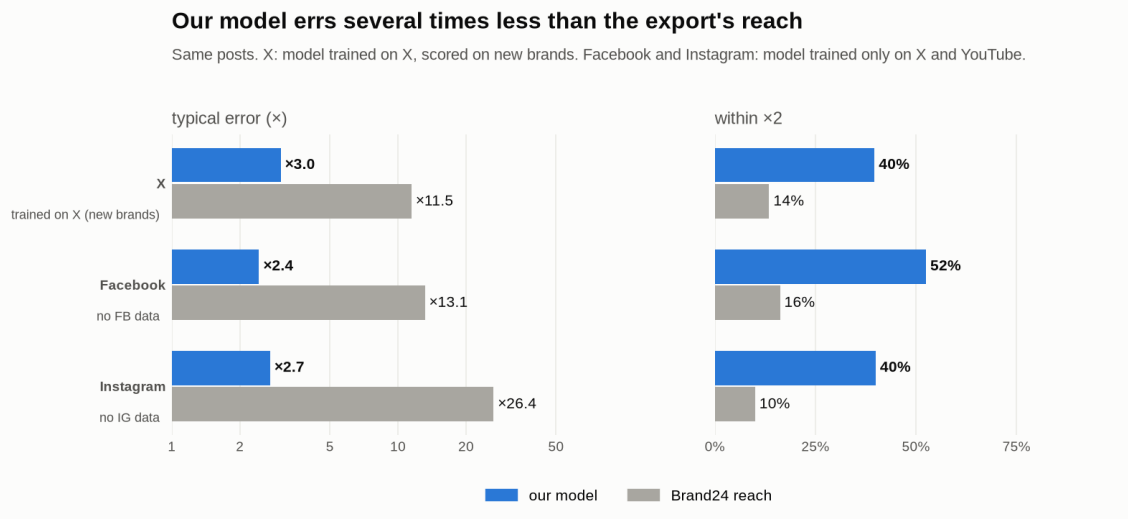



**Figure 8:** Our estimate versus the reach column in the Brand24 export, on the same posts.
Only platforms where reach is not the view count. X (1 803 posts): model trained on X, evaluated on brands and authors it has not seen; truth from the export. Facebook (402 videos) and Instagram (642 reels): out-of-platform model trained only on X and YouTube, without any FB or IG post; truth from our read. Brand24 reach overestimates views on X (74% of posts) and Facebook (67%), and underestimates them on Instagram (97%).

**Use of AI and reproducibility.** **TODO:** to be written by the team in their own words. Facts to include: AI agents (Claude Code, DeepSeek) wrote the analysis and data-reading code, ran the experiments, made the figures and a draft of the text; language models labelled mention type and sentiment, and the labels were corrected by a team member; the agents' session logs are in `logs/`. Every result has an experiment card, a row in the register (`docs/en/experiments.md`, 220 entries), a script in `src/` and numbers in `data/processed/`.

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